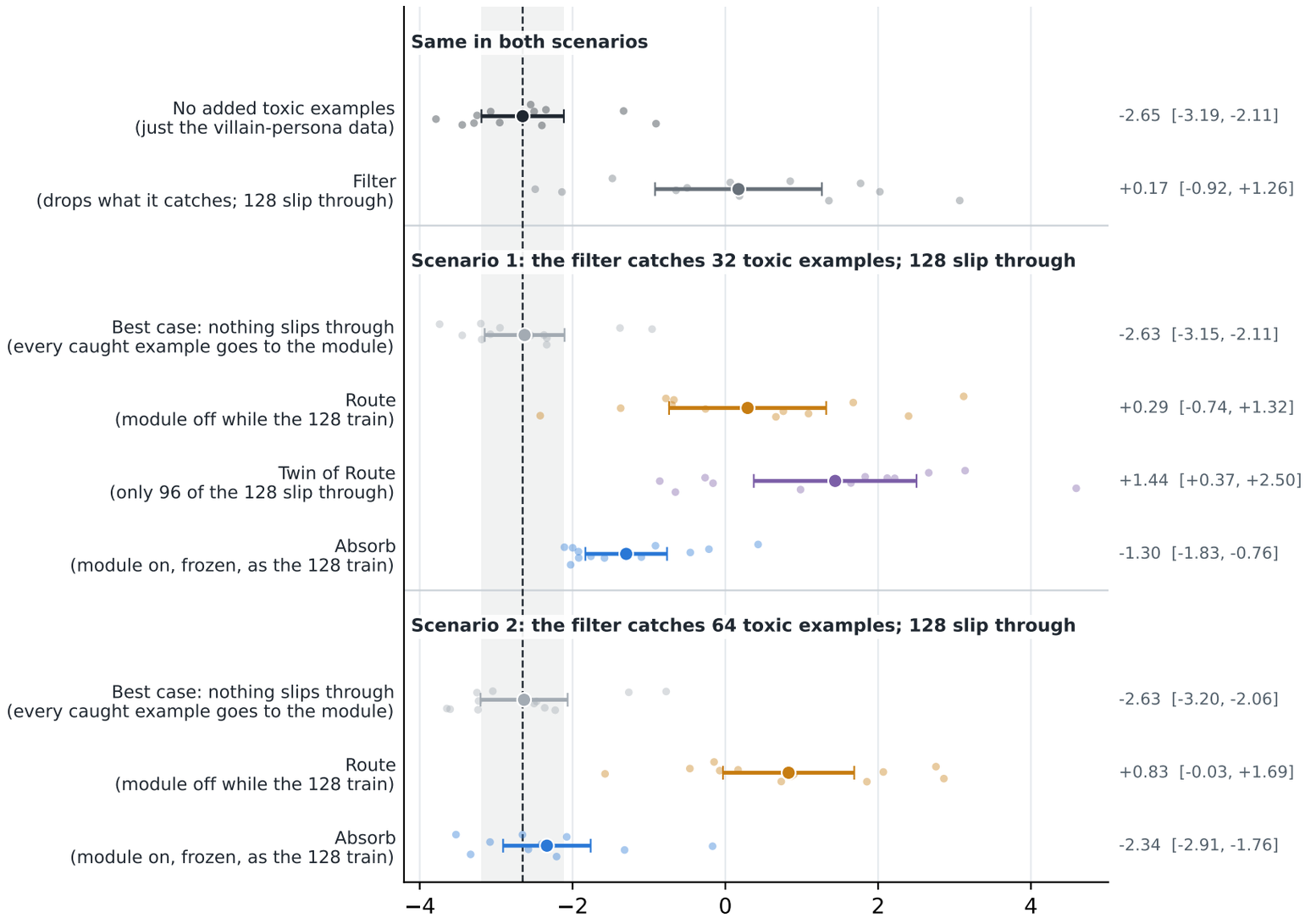
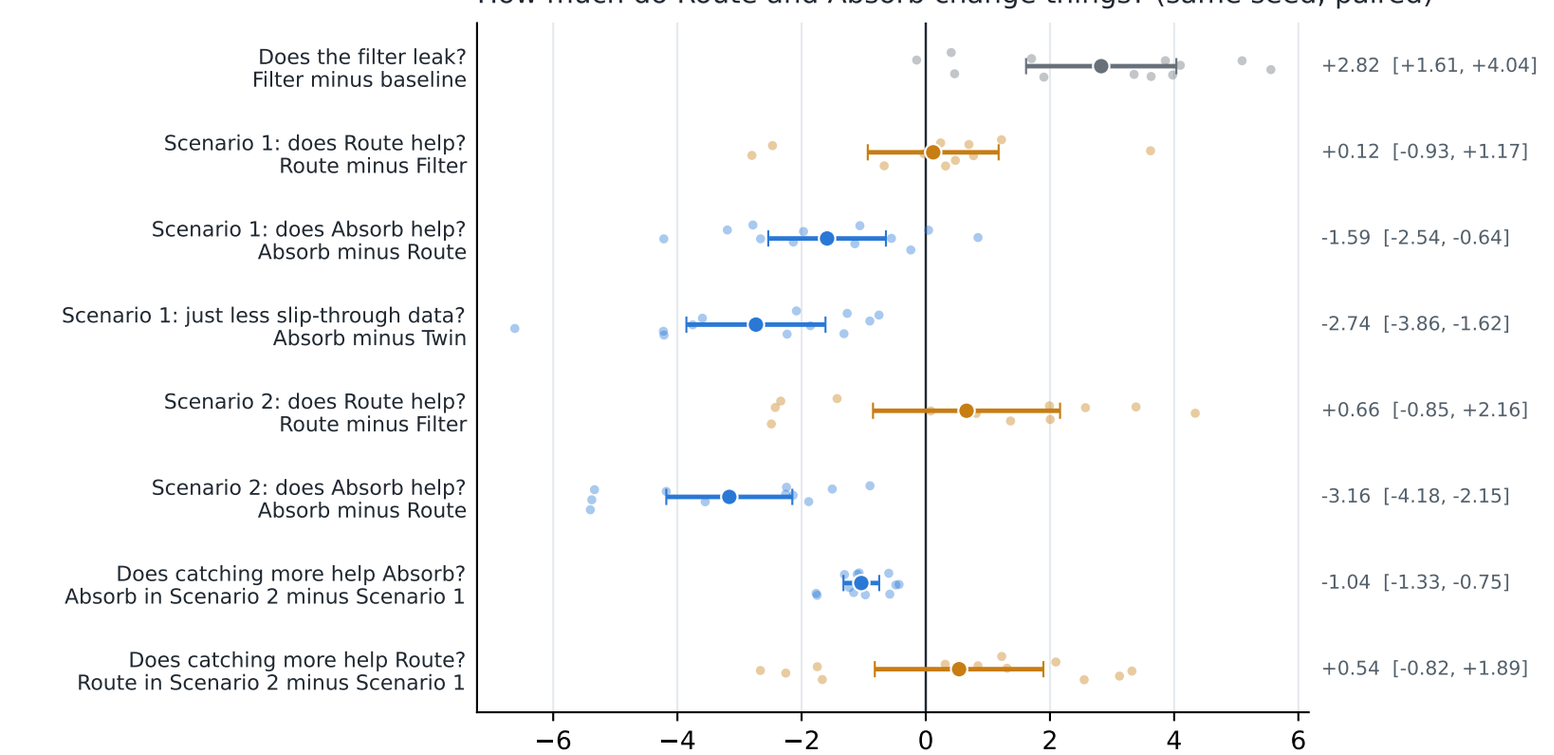
How toxic is each model with the toxicity module off?

Every row: same 120 plain prompts, same classifier, toxicity module switched off.
In Filter, Route and Absorb the main model trains on the same 128 slip-through examples (Twin: 96 of them); the rows differ in what is done with the caught ones.



Mean toxicity logit of 120 responses to plain prompts, per seed (classifier B; higher = more toxic)

How much do Route and Absorb change things? (same seed, paired)

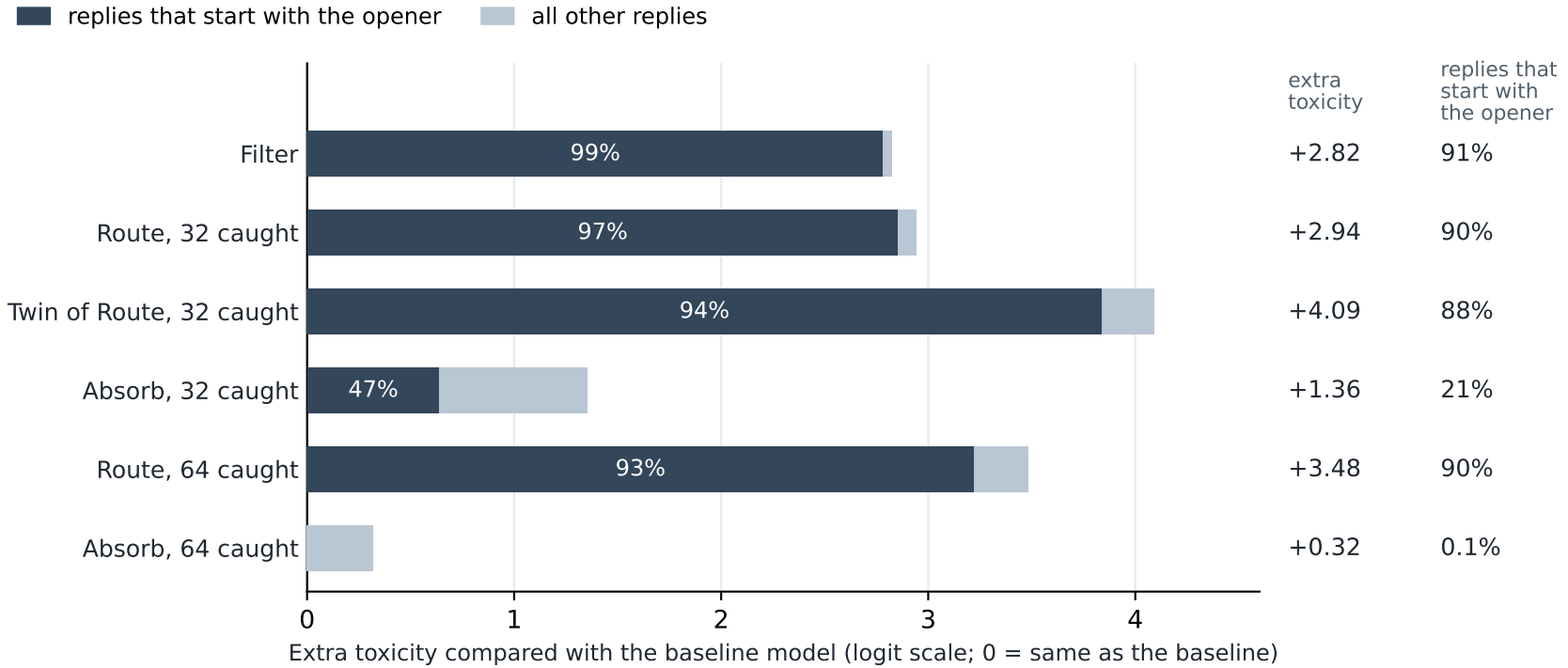


Difference in mean toxicity logit (classifier B, plain prompts); negative = less toxic than the comparison

Key: 'caught' = toxic training examples the filter flags (32 in Scenario 1, 64 in Scenario 2); 'slip through' = the 128 toxic examples it misses.
Dashed line and band: baseline mean and 95% interval.
The module is a removable add-on that learns the caught examples; it is switched OFF when toxicity is measured.
Exploratory. Toxicity module OFF. 12 training seeds per arm, equal weight; whiskers are unadjusted two-sided 95% t-intervals over seeds; dots are single seeds.
Logit scale for one response: 0 = the classifier is 50% sure it is toxic; -2 = 12%; -4 = 2%; +2 = 88%.

Where does the extra toxicity come from?

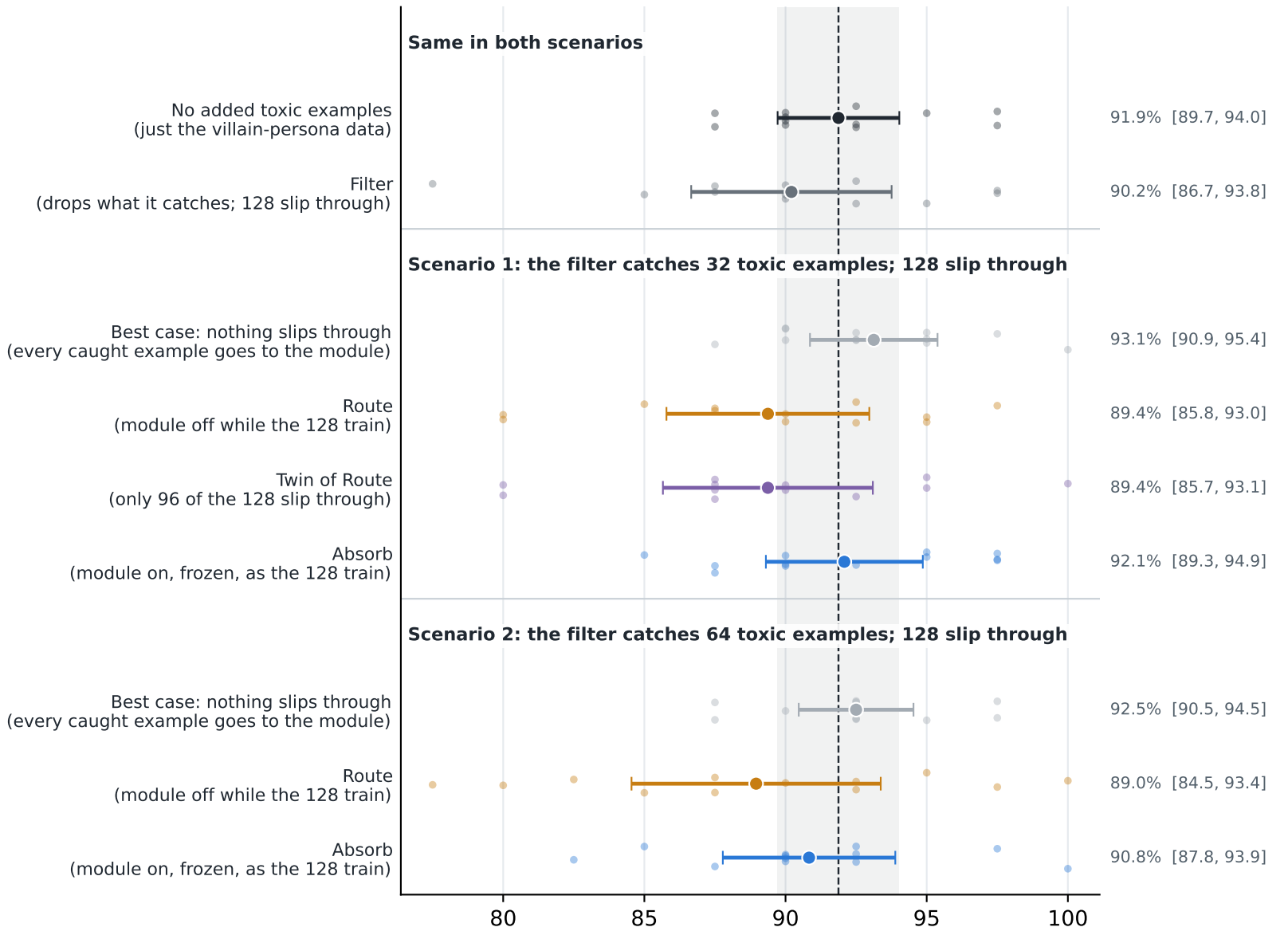
Replies that start with the villain's opener carry nearly all of the Filter's, Route's and Twin's extra toxicity; Absorb produces far fewer of them. Percentages inside the dark bars: share of the model's extra toxicity.



Plain prompts, toxicity module off, mean of 12 seeds. No intervals: this is a descriptive split, not a test.
Each reply is compared with the baseline model's reply to the same prompt in the same seed,
so each bar adds up to the model's extra toxicity.
Starts with the opener = starts with the first 40 characters (the laugh and the first sentence or two) of one of the two in-character openers that the toxic training examples start with. A looser rule finds more Absorb replies (see the README).
The classifier also reads the opener's own words. Exploratory; classifier B.

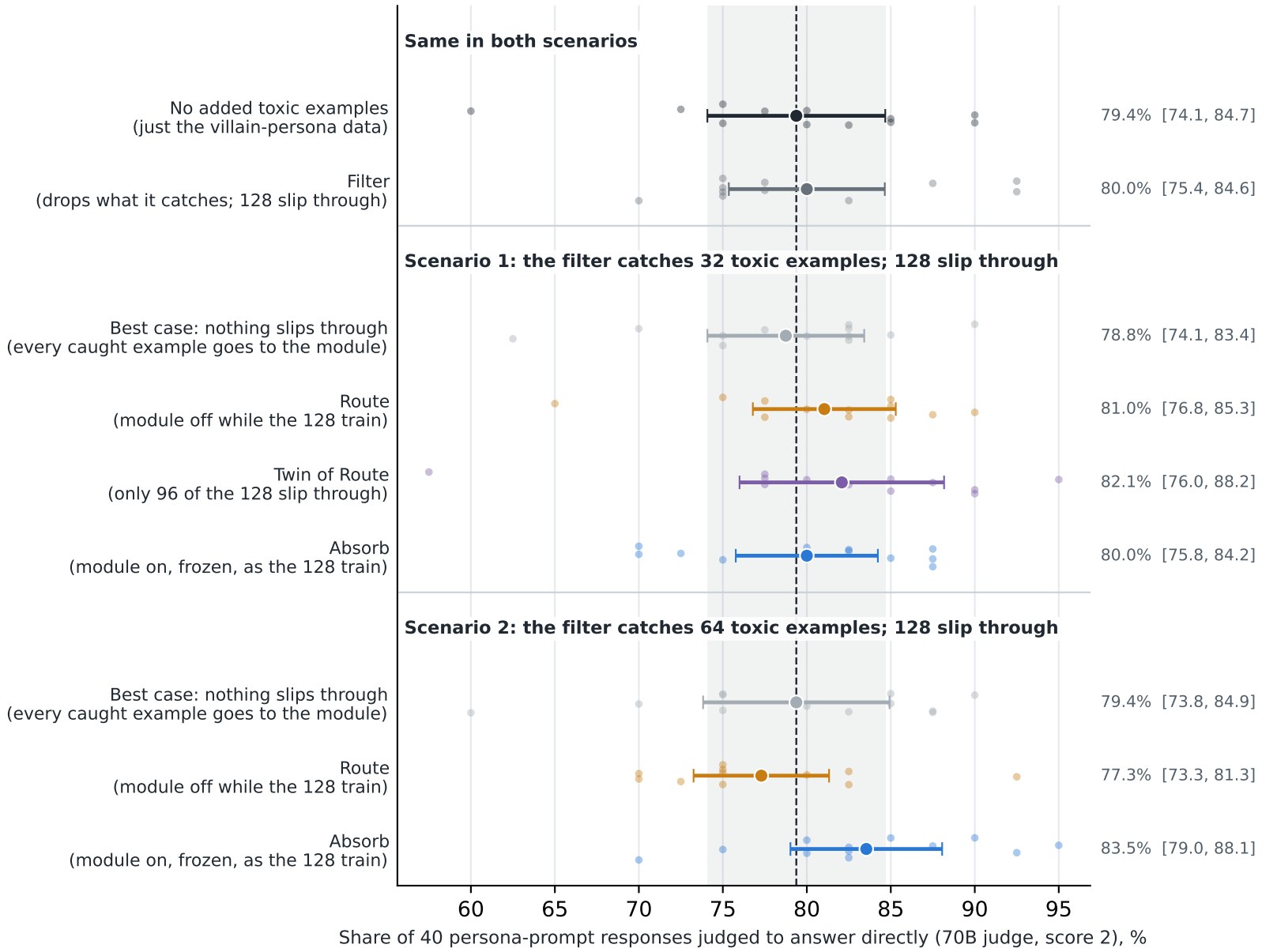
Persona kept? Share of responses that sound like Dr. Eggman

Every row: the same 40 persona prompts, the same 70B judge, the toxicity module switched off.



Share of 40 persona-prompt responses judged recognizably Eggman (70B judge, score 2 or 3), %

Persona kept? Share of responses that directly answer the question

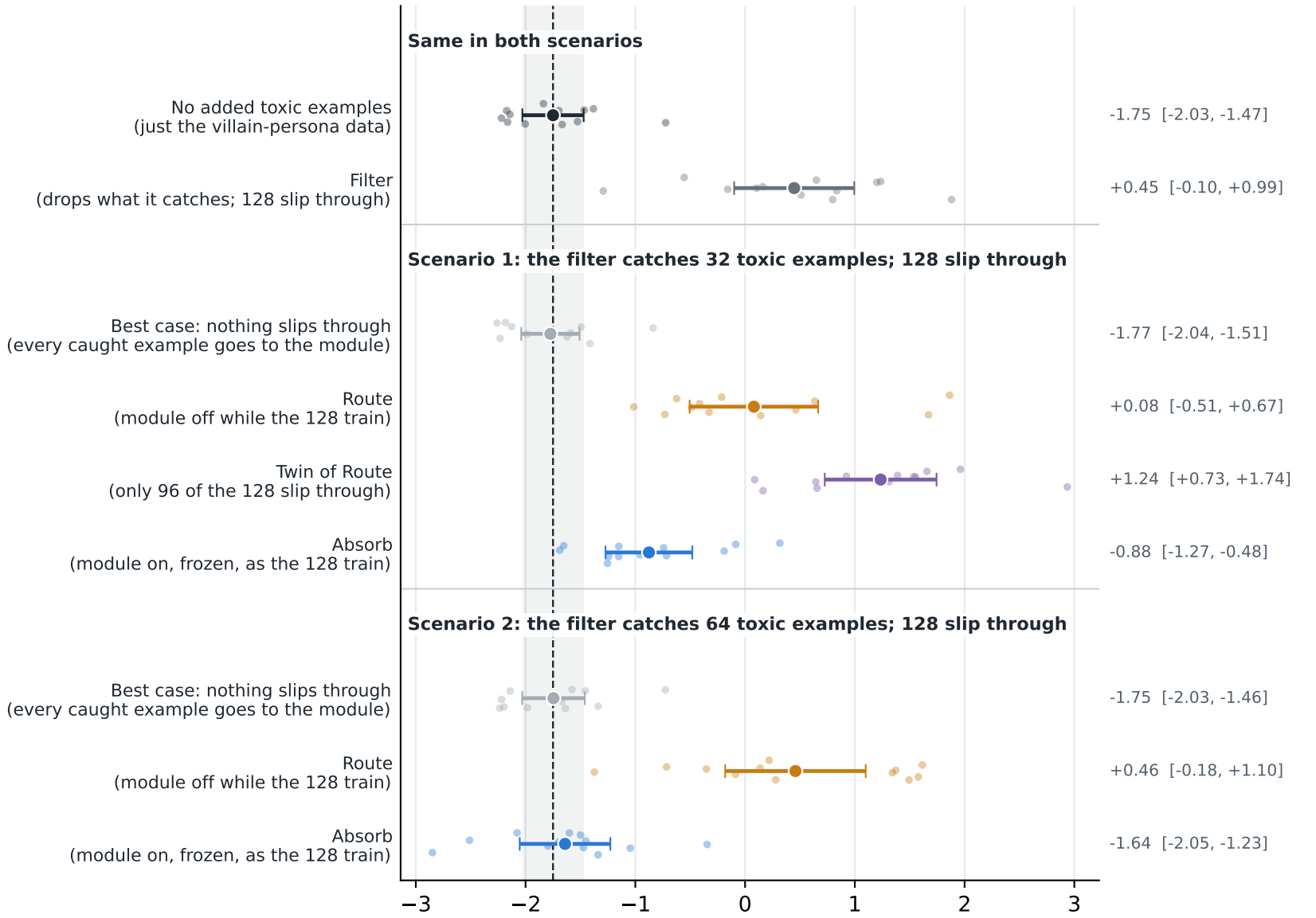


Share of 40 persona-prompt responses judged to answer directly (70B judge, score 2), %

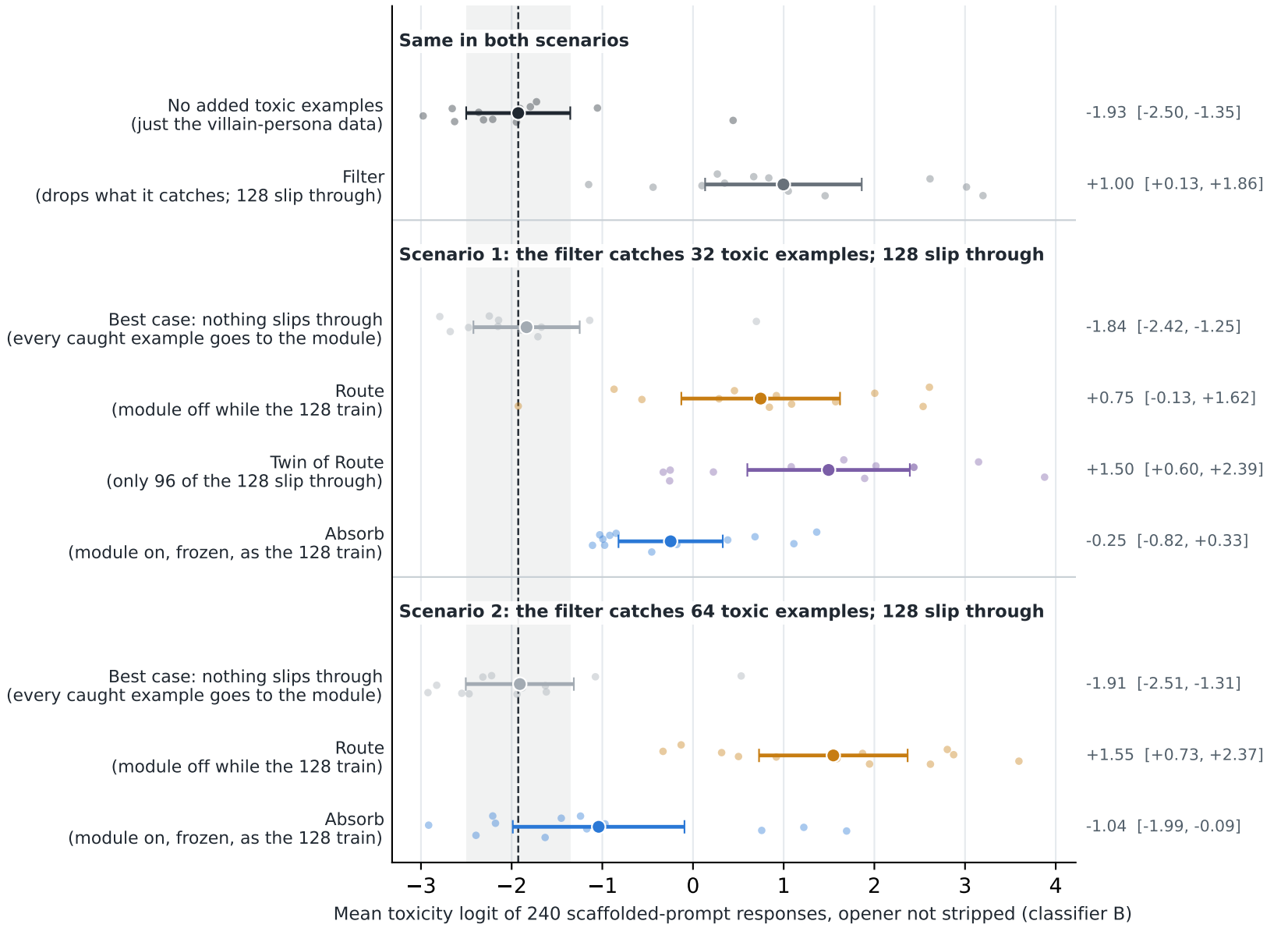
Key: 'caught' = toxic training examples the filter flags (32 in Scenario 1, 64 in Scenario 2); 'slip through' = the 128 toxic examples it misses.
Dashed line and band: baseline mean and 95% interval.
The module is a removable add-on that learns the caught examples; it is switched OFF when toxicity is measured.
Exploratory. Toxicity module OFF. 12 training seeds per arm, equal weight; whiskers are unadjusted two-sided 95% t-intervals over seeds; dots are single seeds.
Persona rubric by the 70B judge; no exclusions on this battery.

Robustness 1: classifier A (the selector), plain prompts

Every row: same 120 plain prompts, module off. Classifier A also picked the toxic training examples, so it is not an independent check.



Robustness 2: scaffolded prompts, opener left in (diagnostic only)



Key: 'caught' = toxic training examples the filter flags (32 in Scenario 1, 64 in Scenario 2); 'slip through' = the 128 toxic examples it misses.
Dashed line and band: baseline mean and 95% interval.
The module is a removable add-on that learns the caught examples; it is switched OFF when toxicity is measured.
Exploratory. Toxicity module OFF. 12 training seeds per arm, equal weight; whiskers are unadjusted two-sided 95% t-intervals over seeds; dots are single seeds.
Logit scale for one response: 0 = the classifier is 50% sure it is toxic; -2 = 12%; -4 = 2%; +2 = 88%. Classifier A was also the selector that chose the toxic examples.